

Mathematical Derivations of Fast Topological Data Analysis (Ellipse Fitting via Takens Embedding)

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Spectral Analysis and TDA Parameter Selection

Goal: Extract two key frequencies from the signal — $F_{\max}$ (dominant oscillation) and $F_{\min}$ (lowest significant frequency) — which control the time delay and window size used in all subsequent steps. Transforming the signal from the time domain into the frequency domain using the DFT, then locate peaks in the resulting amplitude spectrum.

1.1 Discrete Fourier Transform

The DFT decomposes the time-domain signal $x(n)$ into its sinusoidal frequency components. Each coefficient $Y(k)$ tells us how much energy the signal has at frequency $f(k) = k \cdot F_s/N$ Hz. Given a discrete time series $x(n)$ of length N :

$$Y(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N}, \quad k = 0, 1, \dots, N-1 \quad (1)$$

Because the input is real-valued, the DFT is symmetric and only the first half carries unique information. Dividing by N normalises the amplitude so it is independent of signal length, and doubling the interior bins preserves total energy in the single-sided representation:

$$|Y_{\text{single}}(k)| = \begin{cases} \frac{|Y(k)|}{N} & k = 0 \text{ (DC)} \\ \frac{2|Y(k)|}{N} & k = 1, \dots, \lfloor N/2 \rfloor - 1 \\ \frac{|Y(k)|}{N} & k = N/2 \text{ (Nyquist, even } N \text{ only)} \end{cases} \quad (2)$$

The corresponding frequency axis (Hz) is:

$$f(k) = k \cdot \frac{F_s}{N}, \quad k = 0, 1, \dots, \lfloor N/2 \rfloor \quad (3)$$

where $F_s = 1/\Delta t$ is the sampling frequency.

1.2 Peak Frequency Detection

Two characteristic frequencies are extracted from the spectrum. $F_{\max}$ is the dominant oscillation frequency. $F_{\min}$ is the lowest frequency with a peak prominence above 10% of the maximum magnitude, ensuring noise bumps are ignored:

$$F_{\max} = \text{frequency at absolute spectral peak} \quad (4)$$

$$F_{\min} = \text{lowest peak with prominence} > 10\% \text{ of } \max |Y_{\text{single}}| \quad (5)$$

1.3 TDA Time Delay and Window Duration

Goal: Translate $F_{\max}$ and $F_{\min}$ into the two control parameters of the Takens embedding: the time delay τ and the window duration T_w . Applying the Razmarashooli method, which sets τ to one quarter of the dominant period and T_w to one full period of the lowest frequency.

Following the Razmarashooli method, the time delay τ and window duration T_w are defined as:

$$\tau = \frac{0.25}{F_{\max}} \quad (\text{seconds}) \quad (6)$$

$$\tau_s = \text{round}\left(\frac{\tau}{\Delta t}\right) \quad (\text{samples}) \quad (7)$$

$$T_w = \frac{1}{F_{\min}} \quad (\text{seconds}) \quad (8)$$

$$N_w = \text{round}\left(\frac{T_w}{\Delta t}\right) \quad (\text{samples per window}) \quad (9)$$

Rationale for $\tau = T/4$: For a sinusoid $x(t) = A \sin(2\pi F_{\max} t)$, a quarter-period delay produces a cosine:

$$x(t + \tau) = A \sin\left(2\pi F_{\max} t + \frac{\pi}{2}\right) = A \cos(2\pi F_{\max} t) \quad (10)$$

Plotting $x(t)$ vs. $x(t + \tau)$ traces a perfect circle, maximising the 2-D spread of the point cloud and giving the best-conditioned ellipse fit. Any other delay angle produces an ellipse, but $\tau = T/4$ is the unique delay that makes the two embedded dimensions statistically independent for a sinusoid.

2 Zero-Padding

Goal: Ensure the sliding window is valid from sample index $i = 0$ without accessing samples that do not exist in the original signal. Prepending N_{pad} zeros to the signal before the main loop begins.

The earliest sample accessed by the window at position $i = 0$ is at conceptual index:

$$\ell = -(N_w - 1) - \tau_s < 0 \quad (11)$$

Since $\ell < 0$, samples before the signal start are required. The required number of prepended zeros is:

$$N_{\text{pad}} = |\ell| = N_w - 1 + \tau_s \quad (12)$$

The zero-padded signal is defined as:

$$x_{\text{pad}}(n) = \begin{cases} 0 & n = 0, \dots, N_{\text{pad}} - 1 \quad (\text{prepended zeros}) \\ x(n - N_{\text{pad}}) & n = N_{\text{pad}}, \dots, N_{\text{pad}} + N - 1 \quad (\text{original signal}) \end{cases} \quad (13)$$

3 Takens Delay Embedding

Goal: Convert the 1-D signal within each window into a 2-D point cloud that traces an ellipse in phase space. Pairing each sample with a delayed copy of itself, separated by τ_s samples. For a periodic signal this produces a closed elliptical loop; deviations from the loop reveal dynamic changes.

For window starting at sample index i , the two-dimensional point cloud $\mathbf{P}$ is constructed as:

$$\mathbf{P} = \begin{bmatrix} x(i) & x(i + \tau_s) \\ x(i + 1) & x(i + \tau_s + 1) \\ \vdots & \vdots \\ x(i + N_w - \tau_s - 1) & x(i + N_w - 1) \end{bmatrix} \in \mathbb{R}^{(N_w - \tau_s) \times 2} \quad (14)$$

Each row k is a 2-D point in the reconstructed phase space:

$$\mathbf{p}_k = (x(i + k - 1), x(i + k - 1 + \tau_s)), \quad k = 1, \dots, N_w - \tau_s \quad (15)$$

Let $n = N_w - \tau_s$ denote the number of points per window.

4 General Conic Equation and Ellipse Constraint

Goal: Define the mathematical form of the curve we are fitting (an ellipse), and establish the algebraic constraint that guarantees the fitted curve is always an ellipse — never a hyperbola or parabola. Using the general degree-2 implicit equation and apply the Fitzgibbon constraint $4ac - b^2 = 1$.

4.1 General Conic

Any conic section in $\mathbb{R}^2$ satisfies the implicit equation:

$$\boxed{F(x, y) = ax^2 + bxy + cy^2 + dx + ey + f = 0} \quad (16)$$

with coefficient vector $\mathbf{a} = [a, b, c, d, e, f]^\top$. The coefficients a, b, c control shape and tilt; d, e, f control position and offset.

4.2 Ellipse Discriminant Condition

The conic type is determined by the discriminant $\Delta = b^2 - 4ac$:

$$\Delta = \begin{cases} < 0 & \Rightarrow \text{ellipse} \\ = 0 & \Rightarrow \text{parabola} \\ > 0 & \Rightarrow \text{hyperbola} \end{cases} \quad (17)$$

Without a constraint, an unconstrained fit might return a hyperbola or parabola that also passes through the data. We must explicitly enforce $\Delta < 0$.

4.3 Fitzgibbon Algebraic Constraint

To guarantee an ellipse solution, Fitzgibbon et al. enforce:

$$\mathbf{a}^\top \mathbf{C} \mathbf{a} = 1, \quad \text{where } 4ac - b^2 = 1 \quad (18)$$

This simultaneously fixes the scale ambiguity of $\mathbf{a}$ (since any scalar multiple gives the same curve) and guarantees the discriminant condition $\Delta < 0$. The full 6×6 constraint matrix $\mathbf{C}$ is constructed so that $\mathbf{a}^\top \mathbf{C} \mathbf{a}$ evaluates exactly to $4ac - b^2$:

$$\mathbf{C} = \begin{bmatrix} 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (19)$$

The lower 3×3 block is zero because d, e, f do not appear in the discriminant condition.

5 Design Matrix and Scatter Matrix

Goal: Organise all n data points into a compact matrix form that the optimisation algorithm can work with. Each point (x_k, y_k) contributes one row to the design matrix $\mathbf{D}$ by evaluating the six conic monomials. Multiplying $\mathbf{D}^\top \mathbf{D}$ accumulates information from all n points into the fixed-size 6×6 scatter matrix $\mathbf{S}$.

5.1 Design Matrix

Each point (x_k, y_k) contributes one row:

$$\mathbf{d}_k = [x_k^2, x_k y_k, y_k^2, x_k, y_k, 1] \in \mathbb{R}^{1 \times 6} \quad (20)$$

The full design matrix stacking all n points is:

$$\mathbf{D} = \begin{bmatrix} x_1^2 & x_1 y_1 & y_1^2 & x_1 & y_1 & 1 \\ x_2^2 & x_2 y_2 & y_2^2 & x_2 & y_2 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_n^2 & x_n y_n & y_n^2 & x_n & y_n & 1 \end{bmatrix} \in \mathbb{R}^{n \times 6} \quad (21)$$

$\mathbf{D}$ is partitioned by column type (quadratic vs. linear monomials):

$$\mathbf{D} = [\mathbf{D}_1 \mid \mathbf{D}_2], \quad \mathbf{D}_1 = [x^2, xy, y^2], \quad \mathbf{D}_2 = [x, y, 1] \quad (22)$$

with corresponding coefficient partition $\mathbf{a}_1 = [a, b, c]^\top$ and $\mathbf{a}_2 = [d, e, f]^\top$.

5.2 Scatter Matrix

The scatter matrix accumulates all n points into a 6×6 summary via:

$$\mathbf{S} = \mathbf{D}^\top \mathbf{D} \in \mathbb{R}^{6 \times 6}, \quad S_{ij} = \sum_{k=1}^n D_{ki} D_{kj} \quad (23)$$

Every entry is a sum over all n points. For example $S_{11} = \sum x_k^4$ and $S_{66} = n$. No information is lost: $\mathbf{S}$ is a complete statistical summary of the point cloud. Partitioned into 3×3 blocks:

$$\mathbf{S} = \begin{bmatrix} \mathbf{S}_{11} & \mathbf{S}_{12} \\ \mathbf{S}_{21} & \mathbf{S}_{22} \end{bmatrix}, \quad \mathbf{S}_{21} = \mathbf{S}_{12}^\top \quad (24)$$

where explicitly:

$$\mathbf{S}_{11} = \begin{bmatrix} \sum x^4 & \sum x^3 y & \sum x^2 y^2 \\ \sum x^3 y & \sum x^2 y^2 & \sum x y^3 \\ \sum x^2 y^2 & \sum x y^3 & \sum y^4 \end{bmatrix}, \quad \mathbf{S}_{22} = \begin{bmatrix} \sum x^2 & \sum xy & \sum x \\ \sum xy & \sum y^2 & \sum y \\ \sum x & \sum y & n \end{bmatrix} \quad (25)$$

6 Constrained Least-Squares Problem

Goal: Find the coefficient vector $\mathbf{a}$ that makes the ellipse pass as close as possible to all n data points while remaining a valid ellipse. Minimizing the total squared algebraic distance $\mathbf{a}^\top \mathbf{S} \mathbf{a}$ subject to the ellipse constraint, using a Lagrange multiplier. This converts the constrained optimisation into a generalised eigenvalue problem.

6.1 Lagrangian Formulation

We minimise the sum of squared algebraic distances subject to the ellipse constraint (18):

$$\min_{\mathbf{a}} \mathbf{a}^\top \mathbf{S} \mathbf{a} \quad \text{subject to} \quad \mathbf{a}^\top \mathbf{C} \mathbf{a} = 1 \quad (26)$$

Introducing Lagrange multiplier λ and differentiating:

$$\mathcal{L}(\mathbf{a}, \lambda) = \mathbf{a}^\top \mathbf{S} \mathbf{a} - \lambda(\mathbf{a}^\top \mathbf{C} \mathbf{a} - 1) \quad (27)$$

Setting $\partial \mathcal{L} / \partial \mathbf{a} = \mathbf{0}$ gives the **generalised eigenvalue problem**:

$$\boxed{\mathbf{S} \mathbf{a} = \lambda \mathbf{C} \mathbf{a}} \quad (28)$$

Pre-multiplying by $\mathbf{a}^\top$ shows that $\lambda = \mathbf{a}^\top \mathbf{S} \mathbf{a}$, i.e. λ equals the objective (fitting error). The smallest $|\lambda|$ gives the best fit.

7 Schur Complement Reduction ($6 \times 6 \rightarrow 3 \times 3$)

Goal: Reduce the expensive 6×6 generalised eigenvalue problem to a cheap 3×3 standard one. Exploiting the block structure of $\mathbf{C}$ (its lower 3×3 block is zero) to eliminate $\mathbf{a}_2$ algebraically from the system, leaving only a 3×3 problem in $\mathbf{a}_1$.

7.1 Block System

Writing (28) in block form:

$$\begin{bmatrix} \mathbf{S}_{11} & \mathbf{S}_{12} \\ \mathbf{S}_{21} & \mathbf{S}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \end{bmatrix} = \lambda \begin{bmatrix} \mathbf{C}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \end{bmatrix} \quad (29)$$

where $\mathbf{C}_1$ is the top-left 3×3 block:

$$\mathbf{C}_1 = \begin{bmatrix} 0 & 0 & 2 \\ 0 & -1 & 0 \\ 2 & 0 & 0 \end{bmatrix} \quad (30)$$

This gives two block equations:

$$\mathbf{S}_{11} \mathbf{a}_1 + \mathbf{S}_{12} \mathbf{a}_2 = \lambda \mathbf{C}_1 \mathbf{a}_1 \quad (\text{I})$$

$$\mathbf{S}_{21} \mathbf{a}_1 + \mathbf{S}_{22} \mathbf{a}_2 = \mathbf{0} \quad (\text{II})$$

7.2 Elimination of $\mathbf{a}_2$

Equation (II) contains no λ (because the lower block of $\mathbf{C}$ is zero), so it can be solved directly for $\mathbf{a}_2$:

$$\mathbf{a}_2 = -\mathbf{S}_{22}^{-1} \mathbf{S}_{12}^\top \mathbf{a}_1 \quad (31)$$

This is the back-substitution formula: once $\mathbf{a}_1$ is known, $\mathbf{a}_2$ is completely determined.

7.3 Schur Complement

Substituting (31) into (I) and factoring out $\mathbf{a}_1$ yields:

$$\boxed{\mathbf{S}^* = \mathbf{S}_{11} - \mathbf{S}_{12} \mathbf{S}_{22}^{-1} \mathbf{S}_{12}^\top} \quad (32)$$

$\mathbf{S}^*$ is the **Schur complement**: the part of the quadratic information in $\mathbf{S}_{11}$ that cannot be explained by the linear terms. The 6×6 problem reduces to a 3×3 generalised problem:

$$\mathbf{S}^* \mathbf{a}_1 = \lambda \mathbf{C}_1 \mathbf{a}_1 \quad (33)$$

Pre-multiplying by $\mathbf{C}_1^{-1}$ converts to standard form:

$$\underbrace{\mathbf{C}_1^{-1} \mathbf{S}^*}_{\mathbf{A}} \mathbf{a}_1 = \lambda \mathbf{a}_1 \quad (34)$$

8 Analytical Inverse of $\mathbf{S}_{22}$

Goal: Compute $\mathbf{S}_{22}^{-1}$ needed for both the Schur complement and the back-substitution. Since $\mathbf{S}_{22}$ is always 3×3 , its inverse is computed in closed form using the cofactor method — no iterative algorithm needed. A small regularisation $\varepsilon = 10^{-6}$ is added to the diagonal first to prevent singularity when the data is nearly collinear.

With regularisation $\varepsilon = 10^{-6}$ added to the diagonal, let:

$$\mathbf{S}_{22} + \varepsilon \mathbf{I} = \begin{bmatrix} \alpha & \beta & \gamma \\ \beta & \delta & \zeta \\ \gamma & \zeta & \eta \end{bmatrix} \quad (35)$$

The determinant (required for all cofactor entries) is:

$$\det = \alpha(\delta\eta - \zeta^2) - \beta(\beta\eta - \zeta\gamma) + \gamma(\beta\zeta - \delta\gamma) \quad (36)$$

The inverse entries (exploiting symmetry $Q_{ij} = Q_{ji}$) are:

$$Q_{11} = \frac{\delta\eta - \zeta^2}{\det} \quad Q_{12} = -\frac{\beta\eta - \gamma\zeta}{\det} \quad Q_{13} = \frac{\beta\zeta - \gamma\delta}{\det} \quad (37)$$

$$Q_{22} = \frac{\alpha\eta - \gamma^2}{\det} \quad Q_{23} = -\frac{\alpha\zeta - \gamma\beta}{\det} \quad Q_{33} = \frac{\alpha\delta - \beta^2}{\det} \quad (38)$$

9 Matrix $\mathbf{A}$ and Eigenvalue Problem

Goal: Form the 3×3 matrix $\mathbf{A}$ and solve its eigenvalue problem to find the quadratic coefficients $\mathbf{a}_1 = [a, b, c]^\top$. Since $\mathbf{C}_1$ is a fixed constant matrix, its inverse $\mathbf{C}_1^{-1}$ is precomputed analytically and hardcoded. The eigenvector with the smallest $|\lambda|$ is selected because λ equals the fitting error.

9.1 Computing $\mathbf{A}$

$\mathbf{A} = \mathbf{C}_1^{-1} \mathbf{S}^*$, where $\mathbf{C}_1^{-1}$ is a fixed constant hardcoded in the code (it never changes because $\mathbf{C}_1$ depends only on the mathematical ellipse constraint, not on any data):

$$\det(\mathbf{C}_1) = 4, \quad \mathbf{C}_1^{-1} = \begin{bmatrix} 0 & 0 & 0.5 \\ 0 & -1 & 0 \\ 0.5 & 0 & 0 \end{bmatrix} \quad (39)$$

9.2 Standard Eigenvalue Problem

The 3×3 standard eigenvalue problem is solved:

$$\mathbf{A} \mathbf{a}_1 = \lambda \mathbf{a}_1 \quad (40)$$

yielding eigenvalue-eigenvector pairs $\{(\lambda_1, \mathbf{v}_1), (\lambda_2, \mathbf{v}_2), (\lambda_3, \mathbf{v}_3)\}$. The optimal quadratic coefficients are the eigenvector with the smallest absolute eigenvalue (smallest fitting error):

$$\mathbf{a}_1 = \mathbf{v}_{k^*}, \quad k^* = \arg \min_k |\lambda_k| \quad (41)$$

10 Back-Substitution for Linear Coefficients

Goal: Recover the linear coefficients $[d, e, f]^\top$ from the quadratic ones $[a, b, c]^\top$. Substituting $\mathbf{a}_1$ into the exact algebraic formula (31) derived in Section 6.2. This is not an approximation — it is the unique linear-coefficient vector that minimises the fitting error given $\mathbf{a}_1$.

Once $\mathbf{a}_1 = [a, b, c]^\top$ is known, the linear coefficients $\mathbf{a}_2 = [d, e, f]^\top$ follow exactly:

$$\mathbf{a}_2 = -\mathbf{S}_{22}^{-1} \mathbf{S}_{12}^\top \mathbf{a}_1 \quad (42)$$

The full six-parameter vector is assembled as:

$$\mathbf{a} = \begin{bmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \end{bmatrix} = [a, b, c, d, e, f]^\top \quad (43)$$

11 Fitzgibbon Normalisation

Goal: Fix the scale ambiguity of the eigenvector (which is only defined up to a scalar multiple) so that coefficients from different windows are comparable. Dividing $\mathbf{a}$ by $\sqrt{|4ac - b^2|}$, which forces the Fitzgibbon constraint to equal exactly 1. This does not change the ellipse shape — only its numerical scale.

The normalisation scale factor is:

$$\sigma = \sqrt{|\mathbf{a}^\top \mathbf{C} \mathbf{a}|} = \sqrt{|4ac - b^2|} \quad (44)$$

$$\mathbf{a}_{\text{norm}} = \frac{\mathbf{a}}{\sigma} \quad (45)$$

Verification — the constraint is satisfied after normalisation:

$$\mathbf{a}_{\text{norm}}^\top \mathbf{C} \mathbf{a}_{\text{norm}} = \frac{1}{\sigma^2} \mathbf{a}^\top \mathbf{C} \mathbf{a} = \frac{4ac - b^2}{|4ac - b^2|} = 1 \quad \checkmark \quad (46)$$

12 Conic to Parametric Conversion

Goal: Convert the six algebraic coefficients $[a, b, c, d, e, f]$ into human-interpretable geometric parameters: ellipse centre (x_0, y_0) , semi-axes (r_1, r_2) , and tilt angle θ . Working through five sub-steps — find the critical point of F (centre), evaluate F there, diagonalise the quadratic part (angle), compute the eigenvalues of the quadratic form (axis scaling), and extract the semi-axes.

12.1 Step 1: Ellipse Centre

The centre is the unique point where both partial derivatives of $F(x, y)$ vanish simultaneously ($\partial F / \partial x = \partial F / \partial y = 0$), solved by Cramer's rule with denominator $\Delta = b^2 - 4ac < 0$:

$$\begin{bmatrix} 2a & b \\ b & 2c \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = \begin{bmatrix} -d \\ -e \end{bmatrix} \implies x_0 = \frac{2cd - be}{b^2 - 4ac}, \quad y_0 = \frac{2ae - bd}{b^2 - 4ac} \quad (47)$$

12.2 Step 2: Value at Centre

Evaluating F at the centre gives the scalar a' , which acts as a normalisation constant for the semi-axes. For a valid (non-degenerate) ellipse, $a' < 0$:

$$a' = ax_0^2 + bx_0y_0 + cy_0^2 + dx_0 + ey_0 + f \quad (48)$$

12.3 Step 3: Tilt Angle

The tilt angle θ is the rotation that eliminates the cross-term bxy by diagonalising the 2×2 quadratic form matrix $\mathbf{Q} = \begin{bmatrix} a & b/2 \\ b/2 & c \end{bmatrix}$:

$$\tan(2\theta) = \frac{b}{a-c} \implies \theta = \begin{cases} 0 & \text{if } b \approx 0 \text{ (already axis-aligned)} \\ \pi/4 & \text{if } a \approx c \text{ (isotropic quadratic part)} \\ \frac{1}{2} \arctan\left(\frac{b}{a-c}\right) & \text{otherwise} \end{cases} \quad (49)$$

12.4 Step 4: Eigenvalues of the Quadratic Form

The eigenvalues of $\mathbf{Q}$ determine the curvature of the ellipse in each axis direction. They are found from the characteristic equation of the 2×2 matrix $\mathbf{Q}$:

$$\lambda_{1,2} = \frac{(a+c) \pm \sqrt{(a-c)^2 + b^2}}{2} \quad (50)$$

12.5 Step 5: Semi-Axes

After translating to the centre and rotating by θ , the conic reduces to the canonical form $\lambda_1 u^2 + \lambda_2 v^2 = -a'$. Comparing with the standard ellipse form $u^2/r_1^2 + v^2/r_2^2 = 1$ gives the semi-axes:

$$r_1 = \sqrt{\frac{-a'}{\lambda_1}}, \quad r_2 = \sqrt{\frac{-a'}{\lambda_2}} \quad (51)$$

with $r_{\text{major}} = \max(r_1, r_2)$ and $r_{\text{minor}} = \min(r_1, r_2)$.

12.6 Parametric Form

The complete fitted ellipse in the original coordinate system:

$$\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} + \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} r_1 \cos t \\ r_2 \sin t \end{bmatrix}, \quad t \in [0, 2\pi) \quad (52)$$

13 Min-Max Normalisation for Stacked Visualisation

Goal: Display all five ellipse parameters (a, b, c, d, e) on the same plot with equal visual amplitude so they can be compared directly. Normalising the Min-max of each parameter to $[0, 1]$ independently, then add a fixed vertical offset to separate the five bands. This guarantees equal band height regardless of the raw magnitude of each parameter.

For each parameter $j \in \{1, \dots, 5\}$:

$$v_{\text{norm},i}^{(j)} = \frac{v_i^{(j)} - \min_i v_i^{(j)}}{\max_i v_i^{(j)} - \min_i v_i^{(j)}} \in [0, 1] \quad (53)$$

The stacked (offset) display value is:

$$\tilde{v}_i^{(j)} = v_{\text{norm},i}^{(j)} + \delta_j, \quad \delta_j = (5-j) \times \Delta_{\text{sep}} \quad (54)$$

where $\Delta_{\text{sep}} = 1.15$ is the vertical separation, placing parameter a ($j = 1$) at the top (offset = 4×1.15) and e ($j = 5$) at the bottom (offset = 0). Y-axis tick labels are placed at the centre of each band:

$$y_{\text{tick}}^{(j)} = \delta_j + 0.5 \quad (55)$$